Geophys. J. R. astr. Soc. (1980) **60**, 419–433

Palaeomagnetic results from two basic volcanic formations in the western Himalayas and a Phanerozoic polar wandering curve for India

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Received 1979 July 3; in original form 1978 September 19

Summary. We present palaeomagnetic results from the Durgapipal and Rudraprayag formations, which are basic volcanic formations in the Lesser Himalayas of Uttar Pradesh State. NRM measurements and AF demagnetization stability tests were made on specimens cored from oriented block samples collected at representative sites. Mean stable remanent magnetic directions were used for calculating the Virtual Geomagnetic Pole (VGP) positions; where necessary tectonic corrections were applied.

The virtual geomagnetic north poles were found to be located at:

(a) Durgapipal (Permian): $\lambda_p = 10^\circ$ S, $L_p = 42^\circ$ W;
(b) Rudraprayag (Silurian–Devonian): $\lambda_p = 30^\circ$ S, $L_p = 12^\circ$ W.

A new, continuous Phanerozoic apparent polar wandering curve for the Indian subcontinent has been plotted from the available palaeomagnetic data and the VGP positions reported in this paper. As a result, the gap in the Indian palaeomagnetic data from the Lower Carboniferous to the Cambrian has been partially filled. The locations of the pole positions for the two formations on the Phanerozoic polar wandering curve for the Indian subcontinent, have been found to coincide with the stratigraphic ages assigned to them on the basis of rather limited geological and palaeontological evidence.

The Cambrian and Permian poles for the Salt Range in the NW Himalayas and the Permian pole for the Kumaon Himalayas are grouped along with the pole positions of contemporaneous formations of the Peninsular Shield. The palaeomagnetic data thus suggests that the two formations are autochthonous in nature.

Introduction

During the last two decades, a number of palaeomagnetic studies have been carried out on rock formations in the Indian subcontinent. While formations belonging to the Mesozoic era have been studied extensively, only scanty data are available for the Tertiary period, the Palaeozoic era and for some Pre-Cambrian horizons. The data have been summarized and reviewed periodically (Athavale *et al.* 1970; McElhinny 1973; Wensink 1973, 1975;

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Klootwijk 1976 etc.). The palaeomagnetic polar wandering path based on the Indian data so far available shows a large gap between the Late Pre-Cambrian–Cambrian and Upper Carboniferous.

The time span of the missing palaeomagnetic data is in the order of 200 to 250 Myr. As a consequence, the polar wandering curve for India (Athavale *et al.* 1970; McElhinny 1973; Wensink 1973, 1975) is incomplete and the present path over this long gap can be drawn only through interpolation.

The gap in the polar wandering path is due mainly to the absence of suitable formations in the Peninsular Shield for the Lower Carboniferous to Late Cambrian periods. The Palaeozoic sedimentary sequence in the Himalayas, including the Salt Range, is considered to be disturbed and in many cases the precise stratigraphic positions of the formations are not reliably known.

It is important to fill the gap in the Indian palaeomagnetic data for the following reasons:

(a) In order that India may be correctly juxtaposed in the reconstruction of Gondwanaland (McElhinny 1973; Wensink 1973; Klootwijk 1973; Embleton & Valencio 1977, for example).

(b) To verify certain current hypotheses, e.g. Irving (1977) suggests a short life span for Wegener's Pangaea, Crawford (1974a) suggests that Tibet was part of the Indian constituent of Gondwanaland, and Stonley (1974) is of the opinion that Tibet became separated from Gondwanaland during the Triassic.

Palaeomagnetic studies of formations in the Himalayas

Although the Himalayas are separated from the main Indian Shield by the wide and elongated Indo-Gangetic alluvial basin, it has been shown on the basis of geological (Veevers, Jones & Talent 1971), structural (Valdiya 1973) and palaeomagnetic (Athavale 1973) evidence that they form part of the Indian subcontinent and that the Peninsular Shield extends underneath the Indo-Gangetic basin as far as the Himalayas.

Palaeomagnetic data from Himalayan formations can therefore be used to complete the polar wandering curve for India.

SEDIMENTARY FORMATIONS

Sedimentary sequences in the main Himalayas are not suitable for palaeomagnetic studies because of the absence of reliable age data. A thick succession of non-fossiliferous sedimentary rocks forms a major portion of the Palaeozoic sequences of the Lesser Himalayas (e.g., Jutogh, Chandpur, Simla slates, Blaini, Infra-Krol and Krol series), (Srivastava 1976). On the basis of lithostratigraphic, structural or regional tectonic considerations these formations have been variously dated from Lower Palaeozoic to Upper Mesozoic by different workers. In the case of Kashmir, Spiti and the Tethyan sequences, which are fossiliferous, only preliminary classification and zoning on the basis of megafossils has been possible so far and exact lithostratigraphic correlation is still under discussion.

VOLCANICS IN THE HIMALAYAS

The four geotectonic zones of the Himalayas, the Outer Siwalik zone, the Lesser Himalayas, the Central zone and the Tethyan zone, exhibit well-documented volcanic episodes over their entire stretch. In the western region, the volcanic activity is represented by the Mandi-Darla volcanics in Himachal Pradesh, the Panjal volcanics in Kashmir, the Phe volcanics and

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the Dras volcanics in Ladakh, the Karnaprayag volcanics and the Bhimtal (Bhowali) volcanics in the Kumaon Himalayas.

Radiometric dates for these formations are not available. With the exception of the Dras volcanic activity, which is dated as Late Cretaceous–Eocene (Gansser 1964), all other volcanic occurrences are considered to have originated between the Late Pre-Cambrian and Upper Palaeozoic. The ages of individual volcanic suites could be estimated only on the basis of the ages of associated sedimentary formations, which in most cases have no reliable ages. Nevertheless, in view of the relative superiority of volcanic rocks over the sedimentary rocks for palaeomagnetic investigations, we have studied two basic volcanic formations from the western Himalayas. They are:

(a) The Durgapipal exposure of Bhimtal (Bhowali) volcanics in the Nainital District of the Kumaon Himalayas, and

(b) The Rudraprayag volcanics in the Karnaprayag metavolcanic belt in the Garhwal District of the Kumaon Himalayas.

The locations of these volcanic formations are shown in Fig. 1.

Criteria for the selection of the sampling sites

Palaeomagnetic studies of the Panjal volcanics in Kashmir have been attempted by other groups on two previous occasions. The samples were found to be too weakly magnetized for laboratory measurements and for this reason the earlier studies were mainly unsuccessful. Accordingly, we selected only oriented block samples from those sites where a hand sample

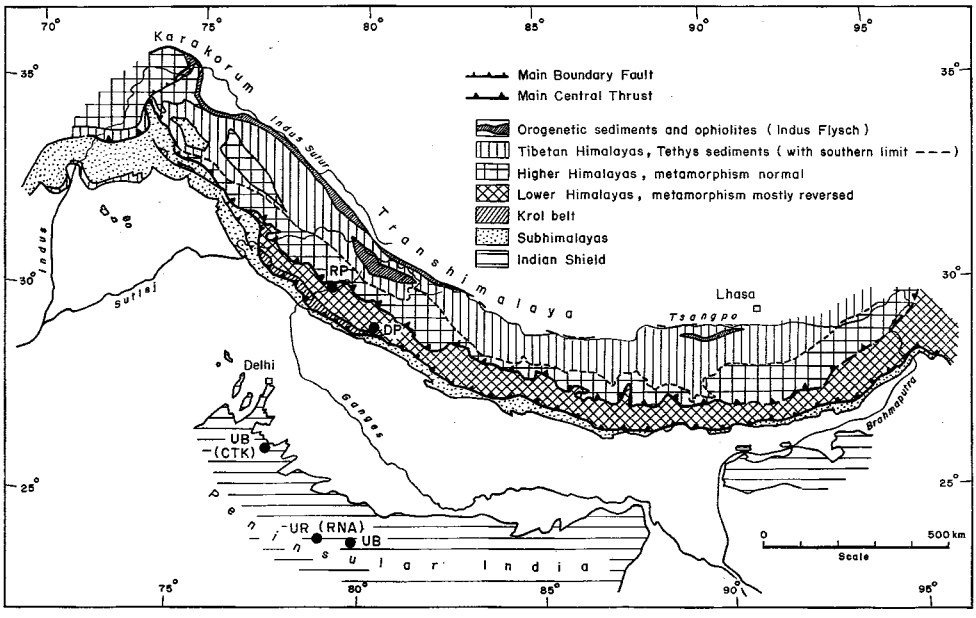


Figure 1. Main geological features of the Himalayas and northern India (after Gansser 1964). The location for formations studied are indicated as: RP, Rudrapayag volcanics; DP, Durgapipal volcanics. Also shown are the sampling sites in the Rewa and Bhander series of the Vindhyan System in the Peninsular Shield. UB (RNA), Upper Bhander (Athavale *et al.* 1972); UB (CTK), Upper Bhander (Klootwijk 1973); UR, Upper Rewa (Athavale *et al.* 1972).

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showed adequate deflection when placed at a distance of about 20 cm from a portable Fluxgate Magnetometer (model 70 California Electronic Manufacturing Company).

Sampling was carried out under the guidance of officers of the Geological Survey of India and only in those areas where the local geology as well as the structural setting had been mapped in some detail.

Geology and sampling details

DURGAPIPAL (BHOWALI–BHIMTAL) VOLCANICS

Samples for palaeomagnetic studies from this locality were collected from an exposure of Bhowali volcanics near Duragapipal. The Bhowali volcanics (also referred to by some investigators as the Bhimtal volcanics) belong to the Deoban Group of formations exposed in the Nainital district of the Kumaon Himalayas. The geology of the area has been described by Raina & Dungrakoti (1975) who report the occurrence of Upper Palaeozoic marine byrozoans in the formation immediately overlying the Bhowali volcanics. According to Shah & Merh (1976), the rocks of the Bhowali–Bhimtal region played an integral part in the depositional history of the basin in which the Krol Group sequence (Blaini–Infrakrol–Krol) was formed. These authors consider the Bhowali volcanics to be of Permo-Carboniferous age.

Twenty-one oriented block samples were collected from five sites in the exposure of Bhowali volcanics at Durgapipal (Fig. 2). The strike and dip were noted at each sampling site.

RUDRAPRAYAG VOLCANICS

The exposures at Rudraprayag, belonging to the Karnaprayag volcanic activity, were sampled for palaeomagnetic studies. The Karnaprayag metavolcanics belong to the Chamoli

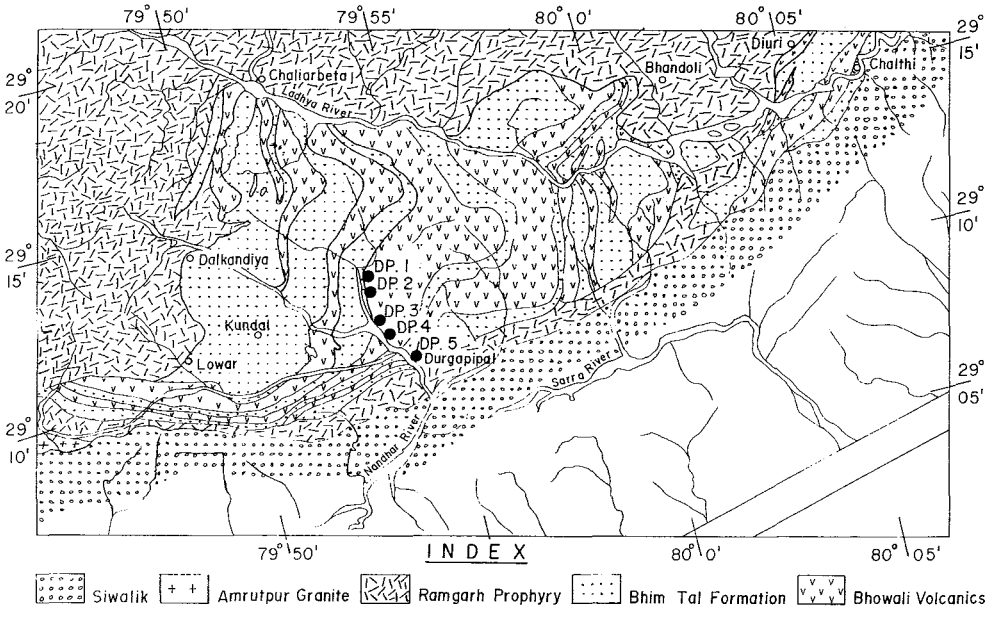


Figure 2. Geological map of the area around Durgapipal, Kumaon Himalayas (after Raina & Dungrakoti 1975), showing palaeomagnetic sampling sites DP1 to DP5.

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Formation of the Garhwal Group, exposed in the Chamoli and Garhwal districts of the Kumaon Himalayas.

Agarwal (1974) has given the following stratigraphic sequence for the Garhwal group:

Gwanagarh Formation
Chamoli Formation
(Dolerites, Metavolcanics, Quartzites)
Lameri and Pipalkoti Formation
(Doleritic Limestones with Slate)
Rudraprayag Formation (Sedimentary)

Agarwal (1974) suggests a normal succession of the various formations and, on the basis of the occurrence of bryozoan fossils, dates the Lameri Formation as Ordovician to Silurian. He further suggests a possible correlation between the Chamoli formation quartzites and the Devonian Muth quartzites of Tethys Himalaya. It is therefore reasonable to assume that the Karnaprayag volcanic group is either Devonian or Silurian.

The exposures of the volcanics at Karnaprayag were found to be weakly magnetized when subjected to field testing with the Fluxgate Magnetometer. The collection was therefore made at Rudraprayag (Fig. 3). Thirty-three oriented block samples were collected from six sites.

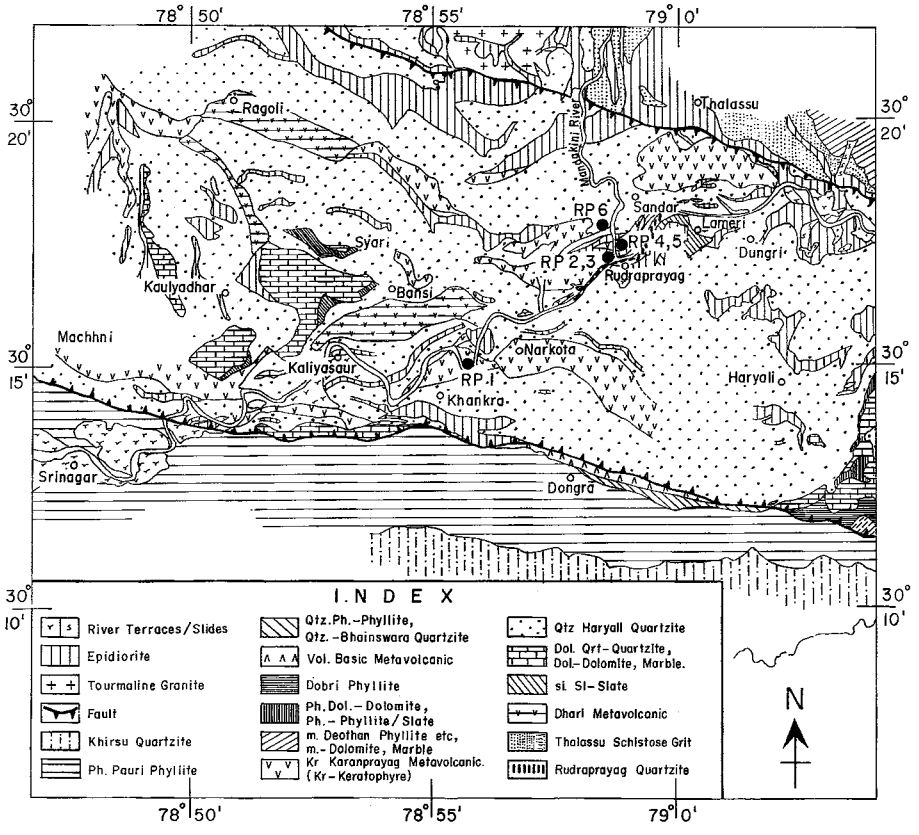


Figure 3. Geological map of area around Rudraprayag, Kumaon Himalayas (after Kumar & Agarwal 1975), showing palaeomagnetic sampling sites RP1 to RP6.

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Laboratory measurements

NRM DIRECTIONS AND MAGNETIC PROPERTIES

Two or three cylindrical cores with a diameter of 2.54 cm were drilled out from each block sample and these cores were cut into specimens of equal height. The remanent magnetic directions were measured on an astatic magnetometer with a sensitivity of up to 3×10^{-7} Oe cm^{-1} deflection at 4 m optical path. The magnetic susceptibilities were measured on an apparatus designed by Likhite & Radhakrishnamurthy (1965).

Durgapipal Area: NRM measurements were carried out on 76 specimens from five sites. The NRM directions show magnetizations in NE and NW quadrants with positive (downward) and negative (upward) dips. The sites DP1, DP3 and DP5 have a magnetization with northerly declination and positive dips, site DP4 a magnetization with northerly declination and negative dip, whereas site DP2 has a magnetization with northerly declination and both positive and negative dips. The grouping of the specimen directions varies from site to site. The site mean NRM directions and other parameters are given in Table 1. The NRM intensity (Jn) shows a variation between 9.0×10^{-5} and 19.9×10^{-4} G. The measured volume susceptibility (χ) of all the specimens varies between 3.0×10^{-5} and 19.9×10^{-3} G Oe^{-1}

Rudraprayag Area: The NRM measurements were carried out on 124 specimens from six sites. The NRM directions of these specimens exhibit a magnetization with northerly declination (NE and NW) and positive (downward) dips. The grouping of these directions varies from site to site. The site mean NRM directions and other parameters are listed in Table 2. The measured NRM intensity (Jn) and volume susceptibility (χ) of these specimens vary between 3.4×10^{-5} and 25.7×10^{-4} G and 12.0 and 71.5×10^{-3} G Oe^{-1} respectively.

AF DEMAGNETIZATION STUDIES

The AF demagnetization studies were carried out on an apparatus similar to that described by Creer (1959). Sixteen pilot specimens from Durgapipal volcanics and 24 pilot specimens from Rudraprayag volcanics were demagnetized in increasing alternating peak fields of 25, 50, 75, 100, 150, 200, 300 and 400 Oe. As already described, these volcanics are rather weakly magnetized, having low remanent coercivity values and showing a wide range in the susceptibility and intensity values. These effects were noticeable in the vector migration and intensity decay measured on successive demagnetizations. A few pilot specimens showed a change in the remanent magnetization vector with increasing demagnetizing fields. These results gave rise to doubts about the performance of our AF demagnetization equipment, particularly in view of the reasonably good grouping of NRM vectors shown at some of the sites.

A special check was made on the performance of the AF demagnetization apparatus while simultaneously the demagnetization characteristics of the Durgapipal and Rudraprayag specimens were studied.

Some specimens from the Durgapipal and Rudraprayag volcanics were given a fresh TRM by heating them up to 670°C and then cooling them in the laboratory field. These specimens, having no multi-component magnetization, were demagnetized along with corresponding virgin specimens from the same block sample. The tumbling device of our AF demagnetization apparatus can take two specimens at a time, enabling one virgin specimen and one heated specimen to be tumbled together in increasing peak-fields. Typical results of these experiments are shown in Figs 4 and 5. The vector migration curves for virgin specimens show a tendency to cluster around certain peak field values.

The optimum peak field values for batch demagnetization of the remaining specimens

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Table 1. Results of palaeomagnetic measurements on samples from Durgapipal.

Site	N	n	NRM D_m	NRM I_m	NRM k	NRM α_{95}	Treatment	*In situ* D_m	*In situ* I_m	After tectonic correction D_m	After tectonic correction I_m	After tectonic correction k	After tectonic correction α_{95}	Pole position λ_p	Pole position L_p
DP1	5	17	47	+51	4	30	100 Oe	12	+44	277	+56	17	17	–	–
DP2	4	17	356	−18	3	43	100 Oe	6	−59	265	−49	10	22	18° S	33° W
DP3	4	20	23	+24	6	29	150 Oe	25	−40	276	−67	9	24	18° S	37° W
DP4	5	18	349	−38	20	14	NRM	349	−38	288	−39	20	14	4° N	39° W
DP5	3	4	328	+21	29	15	100 Oe	319	−20	301	−8	8	29	–	–
Mean			2	+9	3	34		356	−27	284	−26	3	36		
								(k = 3, α_{95} = 34)*							
Mean of DP2, DP3 and DP4										277	−52	25	16	10° S	42° W
														(δ_p = 16, δ_m = 12)	

N = Number of samples.
n = Number of specimens.
D_m = Mean declination east of north.
I_m = Mean inclination positive downwards.
k = Precision parameter.
α_{95} = Radius in the 95 per cent circle of confidence.
λ_p, L_p } Palaeo-coordinates of the Pole.
δ_p, δ_m } Semi axes of oval covariance.

* The k and α_{95} values in brackets are for mean direction before tectonic correction.

Table 2. Results of palaeomagnetic measurements on samples from Rudraprayag.*

Site	N	n	NRM D_m	NRM I_m	NRM k	NRM α_{95}	Treatment	*In situ* D_m	*In situ* I_m	After tectonic correction D_m	After tectonic correction I_m	After tectonic correction k	After tectonic correction α_{95}	Pole position λ_p	Pole position L_p
RP1	5	17	5	+59	2	48	75 Oe	122	+58	266	+37	3	37	31° S	21° W
RP2	5	21	326	+80	23	13	NRM	326	+80	67	+40	23	13	31° S	8° W
RP3	5	21	338	+52	8	22	75 Oe	2	+64	62	+26	36	10	23° S	3° W
RP4	5	22	10	+51	45	9	75 Oe	20	+45	66	+10	49	9	33° S	4° W
RP5	6	26	356	+55	42	9	75 Oe	6	+56	60	+20	46	8	33° S	29° W
RP6	7	17	328	+76	6	22	75 Oe	358	+66	67	+49	33	9	30° S	12° W
Mean			353	+63	30	11		19	+69	59	+40	3	30		
								(k = 12, α_{95} = 17)							
Mean of RP2 to RP6										64	+29	26	12	30° S	12° W
														(δ_p = 7, δ_m = 13)	

* Notation used as in Table 1.

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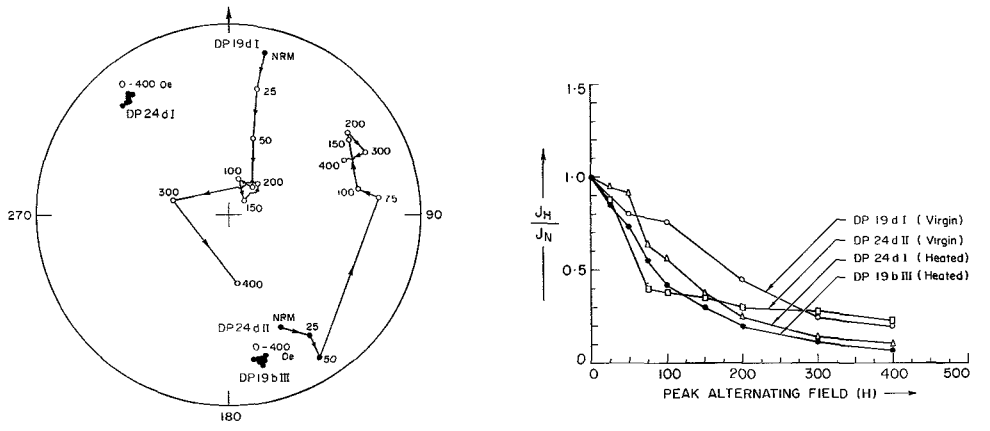


Figure 4. Typical AF demagnetization vector migration plots and intensity decay curves for virgin and heated specimens from Durgapipal volcanics. Numbers in the vector migration plots indicate the peak alternating fields. Solid dots indicate positive dips.

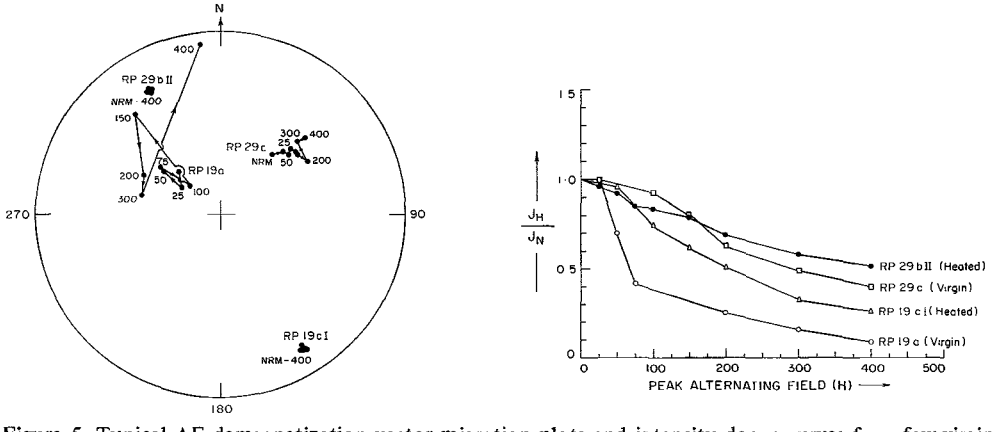


Figure 5. Typical AF demagnetization vector migration plots and intensity decay curves for a few virgin and heated specimens. The numbers in the vector migration plots indicate peak alternating fields. Solid dots indicate positive dips

were selected on the basis of these vector migration diagrams. Tables 1 and 2 give the directions obtained on AF demagnetization for Durgapipal and Rudraprayag sites respectively.

Site mean directions were computed by giving unit weight to the sample and the mean directions for the formations were computed after giving unit weight to each site. The stable remanent directions, obtained after AF cleaning, were corrected for local strike and dip for each site and the corrected directions were used in the calculation of virtual geomagnetic pole positions. The peak fields used and the statistical parameters such as precision (k) and circle of confidence (α_{95}), using statistical analysis as per Fisher (1953), are also given in Tables 1 and 2.

Analysis of results

Figs 6(a) and (b) and 7(a) and (b) respectively show the stable remanent magnetization directions for Durgapipal and Rudraprayag volcanics before and after tectonic correction.

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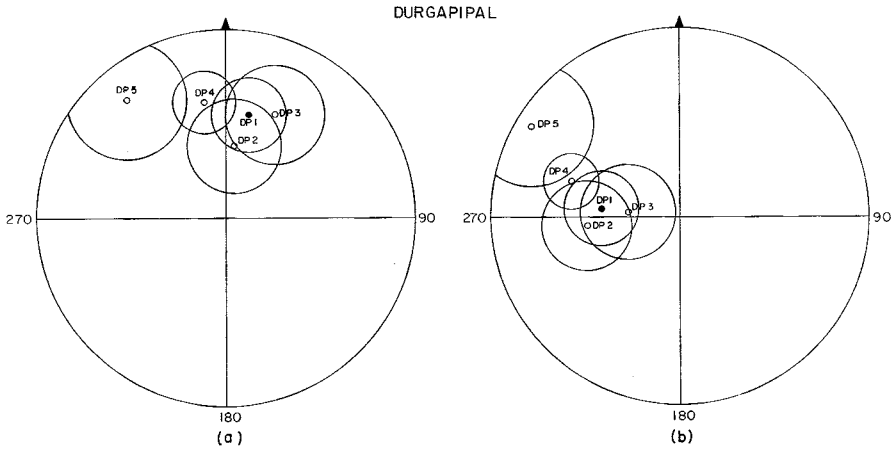


Figure 6. Stereographic plot of the site-mean stable magnetic directions for the Durgapipal volcanics (a) before and (b) after tectonic corrections. The solid circle indicates a positive dip and open circles negative dips.

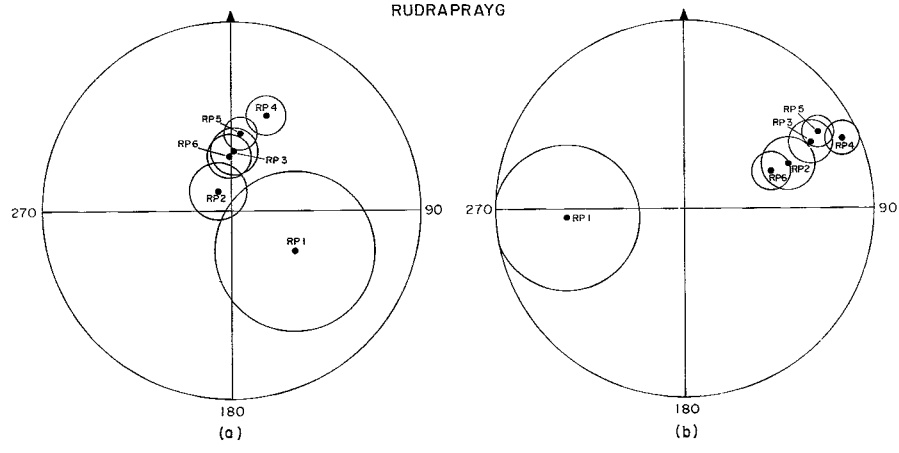


Figure 7. Stereographic plot of the site-mean stable magnetic directions for the Rudraprayag volcanics (a) before and (b) after tectonic corrections.

In the case of Durgapipal volcanics, site DP5 gave shallow negative dip with poor statistics ($k = 8$, $\alpha_{95} = 29$). Four of the seven block samples from this site broke in transit or in coring without yielding any specimens. The remaining three samples yielded only four specimens. Another site, DP1, gave positive dip whereas all the other sites showed negative dips both before and after tectonic correction. The results from these sites were therefore not included in our calculation of the mean direction for the Durgapipal volcanics. In the case of Rudraprayag volcanics, although all the sites were initially grouped together (Fig. 7(a)) on tectonic correction (Fig. 7(b)), site RP1 was found to be located far from the remaining five sites. This site, having a high α_{95} value of 37° was therefore omitted in the calculation of the mean direction.

The precision (k) values for the Durgapipal and Rudraprayag results, showed considerable improvement on being subjected to tectonic correction. The precision parameters for Durgapipal change from 3 to 25 and those for Rudraprayag volcanics change from 12 to 26, as shown in Tables 1 and 2.

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Table 3. Palaeomagnetic data on various Phanerozoic formations from the Indian subcontinent.

Sl. no.	Rock formation	Age	Mean D	Mean I	Polarity	k	α_{95}	λ_m	ψ	VGP positions λ_p	VGP positions L_p	δ_p	δ_m	References
1.	Siwalik Red beds	Mio 1–m	1	+28	N	51.7	17.3	4° N	+1°	72° N	111° W	10.3	19.0	Wensink (1972a)
2.	Deccan Traps (Upper)	60 Myr	345	−44	N	–	10.0	26° S	−15°	42° N	87° W	6.0	13.0	Athavale *et al.* (1970)
3.	Deccan Traps (Lower)	65 Myr	154	+53	R	–	10.0	31° S	−26°	34° N	78° W	7.0	14.0	Athavale *et al.* (1970)
4.	Tirupathi Sandstones	K u	329	−59	N	85.0	4.0	34° S	−31°	28° N	73° W	3.0	6.0	Verma & Pullaiah (1967)
5.	Satyavedu Sandstones	K 1–m	321	−58	N	80.0	4.2	33° S	−39°	26° N	67° W	4.8	6.5	Mital, Verma & Pullaiah (1970)
6.	Rajmahal Traps	100–105 Myr	323	−64	N	60.0	33.5	44° S	−37°	13° N	69° W	4.5	6.0	Radhakrishnamurty (1963)
7.	Sylhet Traps	J–K	332	−59	N	–	7.0	35° S	−28°	16° N	60° W	8.0	10.0	Athavale, Radhakrishnamurty & Sahasrabudhe (1963)
8.	Pachmarhi Sandstones	Tr u	117	+49	R	32.8	4.6	31° S	−63°	10° N	50° W	4.0	6.1	Wensink (1968)
9.	Parsora Sandstones	Tr u	320	−39	N	200.0	5.7	23° S	−40°	30° N	55° W	1.0	3.0	Bhalla & Verma (1969)
10.	Panchet Clay beds	Tr 1	110	+69	R	49.0	6.0	48° S	−70°	8° S	60° W	9.0	10.0	Klootwijk (1974)
11.	Mangli Red beds	Tr 1	100	+63	R	43.3	4.6	44° S	−80°	7° S	56° W	5.7	7.3	Wensink (1968)
12.	Wardha Valley Sandstones	P u–Tr 1	101	+58	R	24.0	6.5	39° S	−79°	4° S	51° W	7.0	9.5	Klootwijk (1975)
13.	Kamthi Red beds	P u	79	+62	R	62.8	2.4	43° S	−101°	21° S	50° W	2.8	3.7	Wensink (1968)
14.	Kamthi Sandstones	P u	306	−50	N	–	8.1	29° S	−54°	18° N	53° W	1.0	3.0	Verma & Bhalla (1968)
15.	Himgir Sandstones	P m	332	−44	N	–	66.0	24° S	−28°	30° N	58° W	3.0	7.0	Verma, Athavale & Agarwal (1973)
16.	Durgapipal Volcanics	P	277	−52	N	25.4	16.0	32° S	−83°	10° S	42° W	10.3	22.0	Present paper
17.	Speckled Sandstones	P 1	92	+47	R	50.0	7.3	34° S	−88°	13° S	43° W	5.1	9.5	Wensink (1975)
18.	Talchir Series	C u	66	+59	R	109.4	2.9	40° S	−114°	31° S	46° W	3.2	2.6	Wensink & Klootwijk (1968)
19.	Rudraprayag Volcanics	S–D	64	+29	R	26.0	12.0	11° S	−116°	30° S	12° W	7.4	13.4	Present paper
20.	Salt Pseudomorph beds	ϵ1–m	218	+36	N	176.0	5.1	25° N	−123°	27° S	33° E	3.0	7.0	Wensink (1972b)
21.	Purple Sandstone	ϵ1	218	+32	N	17.4	11.0	23° N	−142°	28° S	32° E	6.0	12.0	McElhinny (1970)
22.	Upper Bhander Sandstones	Pϵ–ϵ1	49	−19	R	200.0	5.7	12° N	−131°	32° S	19° E	3.1	5.9	Athavale *et al.* (1972)
23.	Upper Bhander Sandstones	Pϵ–ϵ1	208	+10	N	137.5	5.5	10° N	−153°	49° S	34° E	3.0	5.5	Klootwijk (1973)
24.	Upper Rewa Sandstones	Pϵ–ϵ1	32	−37	R	15.0	13.7	24° N	−148°	35° S	42° E	9.6	15.0	Athavale *et al.* (1972)

D = Declination in degrees.
I = Inclination in degrees, positive downwards.
λ_m = Palaeolatitude of Nagpur.
Polarity: N = Normal, R = Reverse.
Rest of the symbols as in Table 1.

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The pole positions obtained for the Durgapipal and Rudraprayag volcanics were plotted along with the other available palaeomagnetic data (Table 3) for the Indian subcontinent. The polar wandering curve thus obtained for the Phanerozoic period is shown in Fig. 9.

Table 3 gives the palaeomagnetic parameters such as mean direction of magnetization, VGP positions, palaeolatitude and palaeorotation etc. for all the formations studied so far, for the Indian subcontinent since the Phanerozoic.

The clockwise rotations are represented in Table 3 by positive angles and anticlockwise rotations by negative angles, measured from 360° azimuth in the case of normal and 180° azimuth in the case of rocks of reversed magnetization.

Table 3 shows that the Indian landmass was rotating anticlockwise in a more or less secular manner. The angles of rotation are less than 90° and the polarity of magnetization can be decided unambiguously by the interpolation of successive results. The mean directions for Speckled Sandstones (Sr. No. 17) and the Talchir Series (Sr. No. 18), however, show high angles of declination of 92° and 66° respectively and positive dips. The ambiguity regarding polarity of magnetization can be resolved on the basis of palaeogeographic and palaeoclimatological evidence.

The Talchir Series are characterized by glacial tillites and boulder beds form the base of the Gondwana System in India. All palaeogeographic and palaeoclimatological evidence (Du Toit 1937) unequivocally confines India to the Southern Hemisphere during the Permian and Carboniferous. The dip angle of +59° obtained for the Talchir formation is therefore indicative of reversed magnetization and consequently the rotation (ψ) had to be taken as $66 - 180 = -114°$, i.e. in the anticlockwise direction for the purpose of calculating the VGP. The angle of rotation for Speckled Sandstones was similarly calculated as $-88°$.

In view of the high angles of rotation indicated by the results for the Permian and Carboniferous, it is difficult to decide from the measured Cambrian and Devonian palaeomagnetic directions whether anticlockwise rotation continued in the past or whether it reversed. This in turn depends on whether India was located in the Northern or the Southern Hemisphere.

We have considered two models for a possible resolution of this ambiguity.

Model 'A' assumes that India was located in shallow Northern Hemisphere latitudes during Cambrian times and that the direction of rotation remained anticlockwise between Cambrian and Carboniferous.

Model 'B' assumes that the directions for Cambrian formations represent shallow Southern Hemisphere latitudes.

Fig. 8(a), (a′), (b) and (b′) depict the palaeolatitudes and palaeorotations, respectively, according to the two models. Model 'B' requires a change in rotations of about 178°, between the Talchir point (300 Myr) and the Rudraprayag point (400 Myr) in about 100 Myr. This would suggest a sharp change in the rotation from Carboniferous to Devonian.

The palaeoclimatic evidence for the Middle Palaeozoic (Devonian–Silurian) period indicates a southern Hemisphere position for India (Du Toit 1937). The Pangaeo reconstruction, based on palaeomagnetic data by Irving (1977), also shows India in the Southern Hemisphere during the Devonian. Therefore, we consider model 'A' to be the more plausible of the two.

The Phanerozoic polar wandering curve for India (Fig. 9) shows a cluster of points in the Cambrian period. Points 22 and 23 are of Bhander Sandstones and point 24 is for Rewa Sandstone belonging to the Vindhyan System of the Peninsular Shield. Points 20 and 21 are for two formations in the Salt Range, which, according to Wadia (1966), is a branch of the 'Tethys Himalaya' tectonic unit. Available geological evidence described earlier (Athavale, Asha Hansraj & Verma 1972) suggests that all these formations are of approximately the

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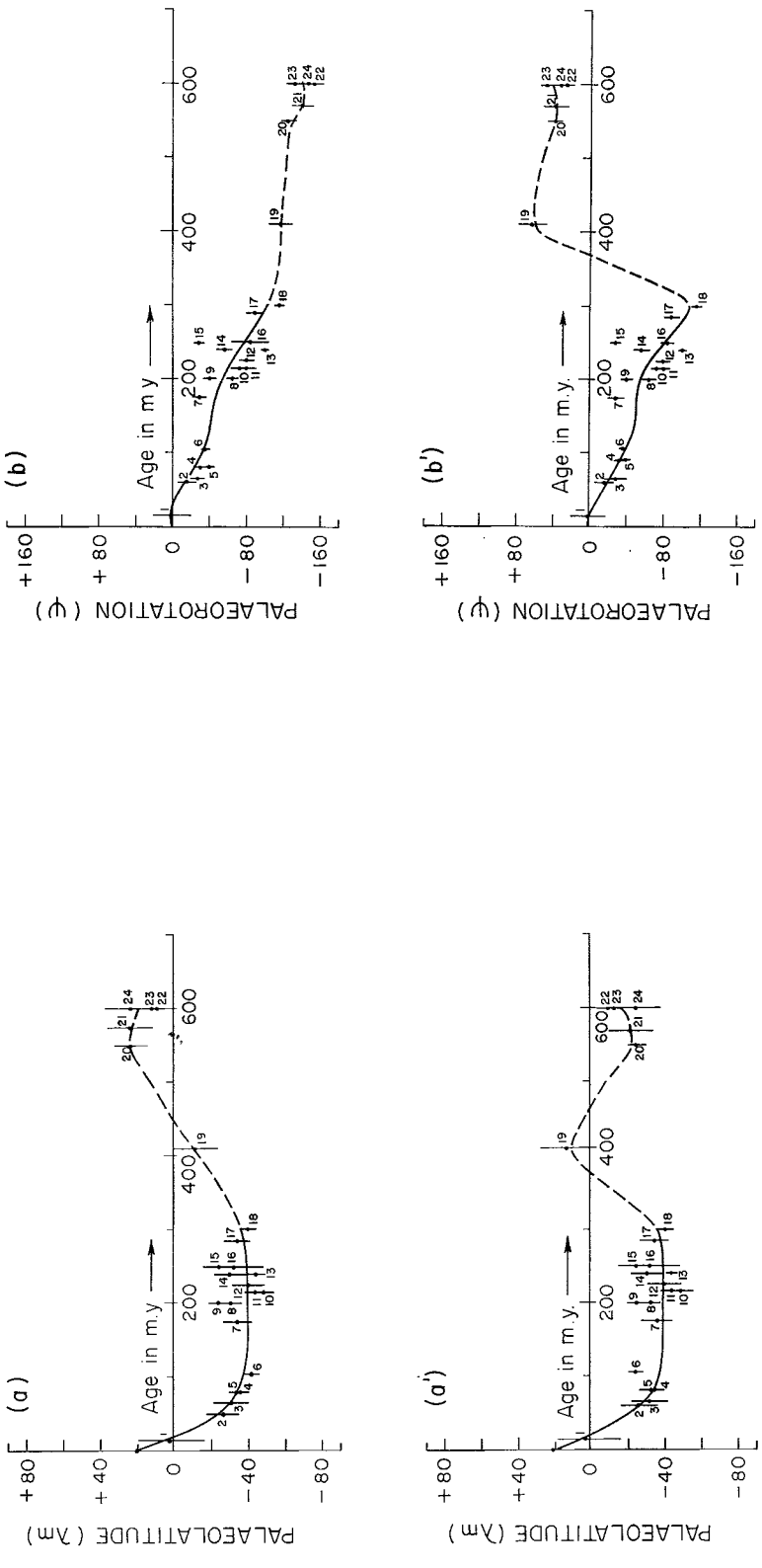


Figure 8. (a) and (a') Palaeolatitudes for India as per models 'A' and 'B' respectively. (b) and (b') Palaeorotations for India as per models 'A' and 'B' respectively. The numbers refer to Sr. Nos. of formations in Table 3.

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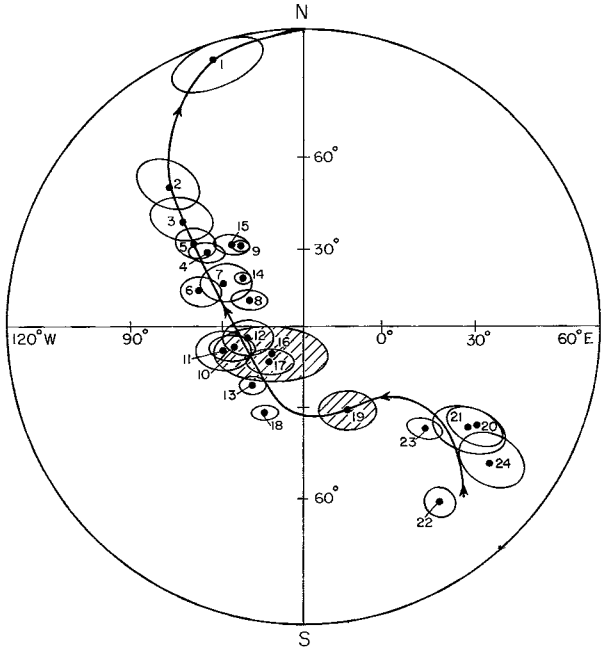


Figure 9. The Phanerozoic palaeomagnetic polar wandering curve for India. The numbers refer to Sr. Nos. of formations listed in Table 3. The two new pole positions reported in this paper are indicated by hatching.

same age. The agreement in the palaeomagnetic data from these groups of formations is considered significant for understanding various theories about inter- and intra-plate tectonics.

Crawford (1974b) has suggested that the agreement between the two poles (nos. 21 and 22) from the Salt Range and those from the Peninsular Shield may be fortuitous and that the palaeomagnetic poles for the Purple Sandstone and the Salt Pseudomorph beds from the Salt Range may be rotated anticlockwise through 75° around a rotation pole located at 33° N, 74° E. According to Crawford, the Salt Range was aligned along the main strike of the Himalayas, but was rotated to occupy the present position as a consequence of tectonic movement taking place some time in the Late Tertiary. However, Wensink (1975) and Klootwijk (1976) have pointed out that the position for Permo-Carboniferous Speckled Sandstone from the Salt Range is located close to those for contemporaneous formations from the Peninsular Shield, although they did not apply the rotation correction suggested by Crawford for the Salt Range palaeomagnetic data. In this paper, we have shown that the Durgapipal volcanics, located in a folded chain (the Lesser Himalayas), also show an agreement with the Permian results from the Salt Range and the Peninsular Shield, without invoking any block rotation.

It is therefore inferred from available palaeomagnetic data that the Peninsular Shield, the Lesser Himalayas of Kumaon and the Himalaya of the Salt Range were constituents of a single tectonic unit (plate), at least until Permian times.

Conclusions

1. Palaeomagnetic studies on the Durgapipal (Bhowali–Bhimtal) and Rudraprayag volcanics in the western Himalayas have given stable remanent magnetic directions. The

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virtual geomagnetic pole positions obtained from these directions have been used to draw a revised Phanerozoic polar wandering curve for India. The gap in palaeomagnetic data for the Middle Palaeozoic has been bridged to a certain extent.

2. The pole positions of these two formations are found to be located on the Indian polar wandering curve at points commensurate with the ages assigned to them on the basis of rather scanty geological and palaeontological data. The Durgapipal pole is located amongst the other Permo-Carboniferous poles. The pole for Rudraprayag volcanics, estimated to be of Silurian to Devonian age, falls in between the Carboniferous and Cambrian poles on the polar wandering curve.

3. The pole positions for the Salt Range (the Tethys Himalayas) and for the Durgapipal and Rudraprayag volcanics (the Kumaon Himalayas) have been calculated after accounting only for the structural dips and strikes at collection sites. The Salt Range Early Cambrian poles for Purple Sandstones and the Salt Pseudomorph beds group together with the pole positions of approximately coeval undisturbed sandstone horizons belonging to the Bhander and Rewa Series of the Vindhyan System of the Peninsular Shield. Similarly, the pole position for Permian Durgapipal volcanics of the Kumaon Himalayas is grouped together with pole positions of coeval formations from the Salt Range and the Peninsular Shield. The palaeomagnetic data therefore suggests an absence of any large scale relative movement between these three geological units of the Indian Plate, at least since Cambrian times. It also suggests that the two Himalayan tectonic units, the Salt Range and the Kumaon Himalayas, are autochthonous in nature.

Acknowledgments

We are grateful to Shri V. S. Krishnaswamy, Director-General, Geological Survey of India, and Shri B. N. Raina, Director, Himalayan Geology Division, G.S.I., for including us in their expeditions to the Himalayas and providing logistics and other help. We are also grateful to Shri B. D. Dungrakoti, Senior Geologist, G.S.I., for his guidance in the field work and for useful discussions. Thanks are due to Dr V. Divakara Rao, N.G.R.I., for useful discussions and to Shri N. T. V. Prasada Rao for his help in the laboratory.

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